Stat 578: Spectral Methods, Network Analysis, and Nonconvex Optimization
HW 1

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Due: 9/25 at 11:59PM

1. Suppose we are given a matrix $\boldsymbol{P}$ of the form $\boldsymbol{P} = \boldsymbol{Z}\boldsymbol{B}\boldsymbol{Z}^\top$, where

   - The matrix $\boldsymbol{Z} \in \mathbb{R}^{n\times K}$ satisfies $0 \le \boldsymbol{Z}_{ik} \le 1$ and $\sum_k \boldsymbol{Z}_{ik} = 1$ for each i.
   - For each k, there is a row i' of $\boldsymbol{Z}$ such that $\boldsymbol{Z}_{i'k} = 1$ and $\boldsymbol{Z}_{i'j} = 0$ for each $k \neq j$ (these are called *pure nodes*)
   - The matrix $\boldsymbol{B} \in [0,1]^{K\times K}$ full rank.

   Suppose that $\boldsymbol{P} = \boldsymbol{U}\boldsymbol{\Lambda}\boldsymbol{U}^\top$ is its eigendecomposition, where $\boldsymbol{U} \in \mathbb{R}^{n\times K}$ are its eigenvectors and $\boldsymbol{\Lambda}$ are its K nonzero eigenvalues.

   (a) Let $\boldsymbol{U}_{\mathsf{pure}}$ denote the rows of $\boldsymbol{U}$ corresponding to the pure nodes. Show that $\boldsymbol{U}_{\mathsf{pure}}$ can be written in terms of $\boldsymbol{Z}$ and $\boldsymbol{B}$.

   (b) Use the expression above to argue that $\boldsymbol{U} = \boldsymbol{Z}\boldsymbol{U}_{\mathsf{pure}}$.

   (c) Show that this implies that each row of $\boldsymbol{U}$ lies on a simplex with corners associated to rows of $\boldsymbol{U}_{\mathsf{pure}}$.

   (d) How does this argument generalize if instead we have that $\boldsymbol{P} = \boldsymbol{\Theta}\boldsymbol{Z}\boldsymbol{B}\boldsymbol{Z}^\top\boldsymbol{\Theta}$, where $\boldsymbol{\Theta}$ is an $n\times n$ diagonal matrix with positive elements?

2. Define the function

$$f(\boldsymbol{x}) := \frac{1}{4}\|\boldsymbol{x}\boldsymbol{x}^\top - \boldsymbol{M}\|_F^2,$$

   and suppose that $\boldsymbol{M}$ is rank one with positive eigenvalue λ. That is, $\boldsymbol{M} = \lambda\boldsymbol{u}\boldsymbol{u}^\top$ for some unit vector $\boldsymbol{u}$.

   (a) Compute $\nabla f(\boldsymbol{x})$ and $\nabla^2 f(\boldsymbol{x})$.

   (b) Show that $\nabla f(\boldsymbol{x}) = 0$ if and only if $\boldsymbol{x}$ is a rescaled eigenvector of $\boldsymbol{M}$.

   (c) Define $\boldsymbol{x}_0 = \sqrt{\lambda}\boldsymbol{u}$, where $\boldsymbol{M} = \lambda\boldsymbol{u}\boldsymbol{u}^\top$. Show that for all $\boldsymbol{x}$ satisfying $\|\boldsymbol{x} - \boldsymbol{x}_0\| \le c\sqrt{\lambda}$ it holds that

$$\nabla^2 f(\boldsymbol{x}) \succ c'\boldsymbol{I}_n,$$

   where c and c' are some sufficiently small universal constants. (As a gauge, this result is possible with $c = \frac{1}{15}, c' = \frac{1}{4}$, but I do not care that you optimize these constants.)

3. In this problem we examine the bounds that result from applying matrix concentration inequalities.

   (a) Suppose that $\boldsymbol{A} \in \{0,1\}^{n\times n}$ is defined via

$$\boldsymbol{A}_{ij} \sim \begin{cases} \text{Bernoulli}(p) & i \le j \\ \boldsymbol{A}_{ji} & i > j. \end{cases}$$

   Show that

$$\|\boldsymbol{A} - p\boldsymbol{1}\boldsymbol{1}^\top\| \le C\sqrt{pn\log(n)}$$

   with high probability, where C is some universal constant. (Update: you may need to assume that $np \ge C\log(n)$ for some sufficiently large constant C.)

(b) Next, suppose that $\boldsymbol{A}$ satisfies

$$\boldsymbol{A} = \lambda \boldsymbol{u}\boldsymbol{u}^\top + \boldsymbol{N},$$

where $\boldsymbol{N}$ is a $\mathsf{GOE}(n)$ random matrix with variance parameter σ^2. Show that

$$\|\boldsymbol{A} - \lambda \boldsymbol{u}\boldsymbol{u}^\top\| \leq 5\sigma$$

with high probability. (Hint: apply Matrix Bernstein with truncation at $\sqrt{\log(n)}$)

(c) Compare the two bounds, and suggest what the optimal bound for $\|\boldsymbol{A} - p\mathbf{1}\mathbf{1}^\top\|$ "ought" to be.